BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Spring 1 Review and Assessment

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Why must a trace table follow executed assignments in order?

2. Where is CPU cache?

A) On a remote server
B) In or near the CPU
C) Inside an optical disc
D) In a backup folder

3. "Secondary storage is needed because RAM permanently keeps every file." Correct this. Explain why.

4. A school uses DVDs for read-only videos and an HDD for a large archive. Correct both claims. Explain why. (a) DVDs are magnetic. (b) HDDs use only flash chips and have no moving parts.

Today's objectives

Complete an independent 45-minute assessment without model answers. Apply systems-software, legal and social-impact knowledge.

Key vocabulary - Read, cover, recall

Operating system
System software managing resources and services.

Utility
Additional system maintenance or protection software.

Data protection
Lawful and secure handling of personal information.

Copyright
Legal protection for original creative work.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Assessment instructions	The Spring 1 assessment covers operating systems, utility software, digital technology impacts, UK legislation, licences, computational thinking and algorithms. Answer every question.
Read the marks and command words	For State, give a clear fact. For Explain, connect a technical feature to its effect. For Evaluate, give a balanced justified judgement.
Show algorithms clearly	Identify inputs, operations and outputs. For flowcharts, label decisions and follow arrows correctly. For trace tables, record variable values after each instruction.
Check before handing in	Read each question once more, show all calculations, use technical vocabulary and answer both sides of comparison or evaluation questions. The worked answers are for use after marking.

I Do - One worked example

Practice only: Explain why storing active programs in RAM helps the computer operate. [2 marks]

1. RAM is used for active programs and working data.

2. RAM holds currently running instructions/data [1].

3. The CPU can access this working information quickly when carrying out instructions [1].

4. Two connected points explain a purpose; this practice example is not part of the timed paper.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Practice only: A school wants to keep private records secure. Explain one benefit of encryption. [2 marks]

2. Practice only: Break down a library fine-calculation program into three smaller tasks. [3 marks]

3. Practice only: Trace total = 5; total = total + 2; print(total). State values and output. [3 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Assessment coding Q1: predict and edit this OCR Exam Reference Language program.

```
unitPrice = 7
quantity = 3
total = unitPrice * quantity
print(total)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Assessment Q2. State TWO purposes of operating systems. [2 marks]

Q3. Assessment Q3. A school uses two programs simultaneously and installs a new printer. Explain how OS memory management and a device driver each help. [4 marks]

Q4. Assessment Q4. A business stores confidential records and needs to reduce the size of old backups. Explain the purposes of encryption and compression utilities. [4 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace table can help show why an algorithm produces an unexpected result. ☐ T ☐ F
2. The Spring 1 assessment covers operating systems, utility software, digital technology impacts, UK legis... ☐ T ☐ F
3. For State, give a clear fact ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Each variable value depends on the state produced by earlier executed instructions. ☐

Q2 B - In or near the CPU ☐

Q3 It keeps programs and files when power is off and usually provides lots of storage space. ☐

Q4 (a) A DVD uses optical storage, read by a laser. (b) An HDD uses magnetic storage on spinning platters. Moving parts and portability affect which is suitable. ☐

Code First - Model

First output: 21 [1]. At quantity = 4: 28 [1].

Possible correct extension:
fee = 2
print(total + fee)

Award [1] for fee/updated addition and [1] for printing the correct updated total of 30. This model assumes the revised quantity is 4.

We Do - Check the model

1. Encryption turns readable data into ciphertext [1], making it difficult to read without the correct key [1]. ☐

2. Input the number of overdue days [1]; calculate fine using a daily rate [1]; display the total [1]. ☐

3. After first assignment total is 5 [1]; after addition total is 7 [1]; output is 7 [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Provide a user interface [1] and manage RAM/application memory [1]. Other correctly described functions: file, peripheral, user management, multitasking. ☐

Q3 The OS allocates RAM to running programs [1], allowing each to hold working data [1]. A device driver enables OS communication with the printer [1] by providing device-specific instructions [1]. ☐

Q4 Encryption converts plaintext to ciphertext [1], helping keep data unreadable without a key [1]. Compression reduces data size [1], saving storage space for backups [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: